

Linux 장애 분석 및 트러블슈팅 보고서

OOM Crash · CPU Latency · Deadlock

실험 환경	Ubuntu 22.04 / aarch64 / UTM
서비스 계정	agent - admin (non - root)
대상 프로그램	agent - leak - app - arm64
서비스 포트	15034
최종 정상 설정	MEMORY_LIMIT=512MB / CPU_MAX_OCCUPY=40% / MULTI_THREAD_ENABLE

[Bug] OOM Crash - 메모리 증가 후 MemoryGuard에 의한 프로세스 종료

1. Description (현상 설명)

MEMORY_LIMIT=256MB 조건에서 MemoryWorker 실행 후 힙 사용량과 프로세스 RSS가 지속적으로 증가했다. 프로그램 내부 힙이 275MB에 도달하자 MemoryGuard가 제한 초과를 감지했고 프로세스가 SELF - TERMINATED 상태로 종료되었다.

2. Evidence & Logs (증거 자료)

외부 관측용 monitor - leak.sh 로그에서 동일 자식 프로세스의 RSS가 약 47.8MB에서 266.7MB까지 지속적으로 증가했다. 프로그램 로그에서는 Current Heap이 275MB에 도달한 직후 Memory limit exceeded 및 SELF - TERMINATED 메시지가 확인되었다.

```
agent-admin@kimseyoan:~$ tail -n 15 /var/log/agent-app/monitor-leak.log
[2026-09-19 10:40:46] PID:129166 CPU:0.5% MEM_RSS:0.9MB
[2026-09-19 10:40:49] PID:129166 CPU:0.4% MEM_RSS:0.9MB
[2026-09-19 10:40:52] PID:129166 CPU:0.3% MEM_RSS:0.9MB
[2026-09-19 10:40:55] PID:129166 CPU:0.3% MEM_RSS:0.9MB
[2026-09-19 10:40:58] PID:129166 CPU:0.3% MEM_RSS:0.9MB
[2026-09-19 10:53:18] PID:129482 CPU:0.8% MEM_RSS:47.8MB
[2026-09-19 10:53:21] PID:129482 CPU:0.6% MEM_RSS:79.8MB
[2026-09-19 10:53:24] PID:129482 CPU:0.5% MEM_RSS:108.4MB
[2026-09-19 10:53:27] PID:129482 CPU:0.4% MEM_RSS:141.7MB
[2026-09-19 10:53:30] PID:129482 CPU:0.4% MEM_RSS:166.7MB
[2026-09-19 10:53:33] PID:129482 CPU:0.4% MEM_RSS:191.7MB
[2026-09-19 10:53:36] PID:129482 CPU:0.3% MEM_RSS:216.7MB
[2026-09-19 10:53:39] PID:129482 CPU:0.3% MEM_RSS:241.7MB
[2026-09-19 10:53:42] PID:129482 CPU:0.3% MEM_RSS:266.7MB
```

그림 1. monitor - leak.sh 관측: 동일 PID의 MEM_RSS가 47.8MB → 266.7MB로 지속 증가

```
2026-09-19 10:53:36,620 [INFO] [MemoryWorker] Current Heap: 200MB
2026-09-19 10:53:39,641 [INFO] [MemoryWorker] Current Heap: 225MB
2026-09-19 10:53:42,662 [INFO] [MemoryWorker] Current Heap: 250MB
2026-09-19 10:53:45,686 [INFO] [MemoryWorker] Current Heap: 275MB
2026-09-19 10:53:45,686 [CRITICAL] [MemoryGuard] Memory limit exceeded (275MB >= 256MB) / (Recommend Over 256MB)
2026-09-19 10:53:45,686 [CRITICAL] [MemoryGuard] Self-terminating process 129482 to prevent system instability.

>>> [SYSTEM] SELF-TERMINATED (Memory Limit Exceeded) <<<
Killed
```

그림 2. MEMORY_LIMIT=256MB: Current Heap 275MB에서 MemoryGuard가 제한 초과를 감지하고 프로세스 종료

3. Root Cause Analysis (원인 분석)

MemoryWorker가 메모리를 계속 확보하면서 프로세스의 실제 상주 메모리(RSS)가 증가했다. 설정된 256MB 한계보다 힙 사용량이 커지자 애플리케이션 내부 MemoryGuard가 보호 동작을 수행한 것이 직접적인 종료 원인이다. 이는 커널 OOM Killer가 동작했다는 증거와는 구분해야 하며, 본 실습 로그에서 확인되는 직접 원인은 애플리케이션의 MemoryGuard이다.

4. Workaround & Verification (조치 및 검증)

최종적으로 MEMORY_LIMIT를 512MB로 조정했다. 정상 설정(512MB / 40% / false)에서는 힙이 525MB에 도달했을 때 프로세스를 종료하지 않고 cleanup을 수행했으며, Memory Cache Flushed 및 MEMORY RECOVERED 이후 작업이 계속 진행되었다. 256MB 조건에서는 약 33초 후 종료되었지만 최종 정상 설정에서는 1분 이상 실행되며 메모리 회수 후에도 생존하는 것을 확인했다.

```
2026-09-19 14:35:08,535 [INFO] [Thread-C] Resumed. Calculating... (100%)
2026-09-19 14:35:08,586 [INFO] [Scheduler] All tasks completed.
2026-09-19 14:35:08,591 [INFO] [MemoryWorker] Current Heap: 25MB
2026-09-19 14:35:08,591 [INFO] [CpuWorker] Started. Maximum CPU Limit: 40%
2026-09-19 14:35:08,592 [INFO] [CpuWorker] Current Load: 5.00%
2026-09-19 14:35:11,614 [INFO] [MemoryWorker] Current Heap: 50MB
2026-09-19 14:35:11,703 [INFO] [CpuWorker] Current Load: 14.05%
2026-09-19 14:35:14,635 [INFO] [MemoryWorker] Current Heap: 75MB
2026-09-19 14:35:14,822 [INFO] [CpuWorker] Current Load: 18.91%
2026-09-19 14:35:17,684 [INFO] [MemoryWorker] Current Heap: 100MB
2026-09-19 14:35:17,943 [INFO] [CpuWorker] Current Load: 19.21%
2026-09-19 14:35:20,714 [INFO] [MemoryWorker] Current Heap: 125MB
2026-09-19 14:35:21,069 [INFO] [CpuWorker] Current Load: 20.94%
2026-09-19 14:35:23,744 [INFO] [MemoryWorker] Current Heap: 150MB
2026-09-19 14:35:24,190 [INFO] [CpuWorker] Current Load: 27.68%
2026-09-19 14:35:26,777 [INFO] [MemoryWorker] Current Heap: 175MB
2026-09-19 14:35:27,314 [INFO] [CpuWorker] Current Load: 35.08%
2026-09-19 14:35:29,804 [INFO] [MemoryWorker] Current Heap: 200MB
2026-09-19 14:35:30,435 [INFO] [CpuWorker] Current Load: 37.04%
2026-09-19 14:35:32,550 [INFO] [CpuWorker] Peak reached (40.00%). Starting cooldown...
2026-09-19 14:35:32,835 [INFO] [MemoryWorker] Current Heap: 225MB
2026-09-19 14:35:33,556 [INFO] [CpuWorker] Current Load: 40.00%
2026-09-19 14:35:35,862 [INFO] [MemoryWorker] Current Heap: 250MB
2026-09-19 14:35:36,680 [INFO] [CpuWorker] Current Load: 33.35%
2026-09-19 14:35:38,891 [INFO] [MemoryWorker] Current Heap: 275MB
2026-09-19 14:35:39,800 [INFO] [CpuWorker] Current Load: 29.00%
2026-09-19 14:35:41,915 [INFO] [MemoryWorker] Current Heap: 300MB
2026-09-19 14:35:42,912 [INFO] [CpuWorker] Current Load: 27.79%
2026-09-19 14:35:44,943 [INFO] [MemoryWorker] Current Heap: 325MB
2026-09-19 14:35:46,037 [INFO] [CpuWorker] Current Load: 24.78%
2026-09-19 14:35:47,973 [INFO] [MemoryWorker] Current Heap: 350MB
2026-09-19 14:35:49,164 [INFO] [CpuWorker] Current Load: 22.57%
2026-09-19 14:35:51,001 [INFO] [MemoryWorker] Current Heap: 375MB
2026-09-19 14:35:52,290 [INFO] [CpuWorker] Current Load: 18.31%
2026-09-19 14:35:54,028 [INFO] [MemoryWorker] Current Heap: 400MB
2026-09-19 14:35:55,414 [INFO] [CpuWorker] Current Load: 12.79%
2026-09-19 14:35:57,040 [INFO] [MemoryWorker] Current Heap: 425MB
2026-09-19 14:35:57,531 [INFO] [CpuWorker] Cooldown complete (5.00%). Resuming load increase...
2026-09-19 14:35:58,541 [INFO] [CpuWorker] Current Load: 5.00%
2026-09-19 14:36:00,062 [INFO] [MemoryWorker] Current Heap: 450MB
2026-09-19 14:36:01,666 [INFO] [CpuWorker] Current Load: 6.99%
2026-09-19 14:36:03,092 [INFO] [MemoryWorker] Current Heap: 475MB
2026-09-19 14:36:04,790 [INFO] [CpuWorker] Current Load: 13.86%
2026-09-19 14:36:06,118 [INFO] [MemoryWorker] Current Heap: 500MB
2026-09-19 14:36:07,913 [INFO] [CpuWorker] Current Load: 16.17%
2026-09-19 14:36:09,144 [INFO] [MemoryWorker] Current Heap: 525MB
2026-09-19 14:36:09,144 [WARNING] [MemoryWorker] Memory Usage Reached Limit (525MB). Starting cleanup...
2026-09-19 14:36:09,160 [INFO] [System] Memory Cache Flushed. Process Stabilized.

>>> [SYSTEM] MEMORY RECOVERED (Cache Cleared) <<<

2026-09-19 14:36:11,039 [INFO] [CpuWorker] Current Load: 25.87%
2026-09-19 14:36:14,165 [INFO] [CpuWorker] Current Load: 34.27%
2026-09-19 14:36:14,198 [INFO] [MemoryWorker] Current Heap: 25MB
2026-09-19 14:36:16,282 [INFO] [CpuWorker] Peak reached (40.00%). Starting cooldown...
2026-09-19 14:36:17,225 [INFO] [MemoryWorker] Current Heap: 50MB
```

그림 3. MEMORY_LIMIT=512MB 정상 설정: 525MB 도달 후 cleanup 및 MEMORY RECOVERED, 이후 작업 지속

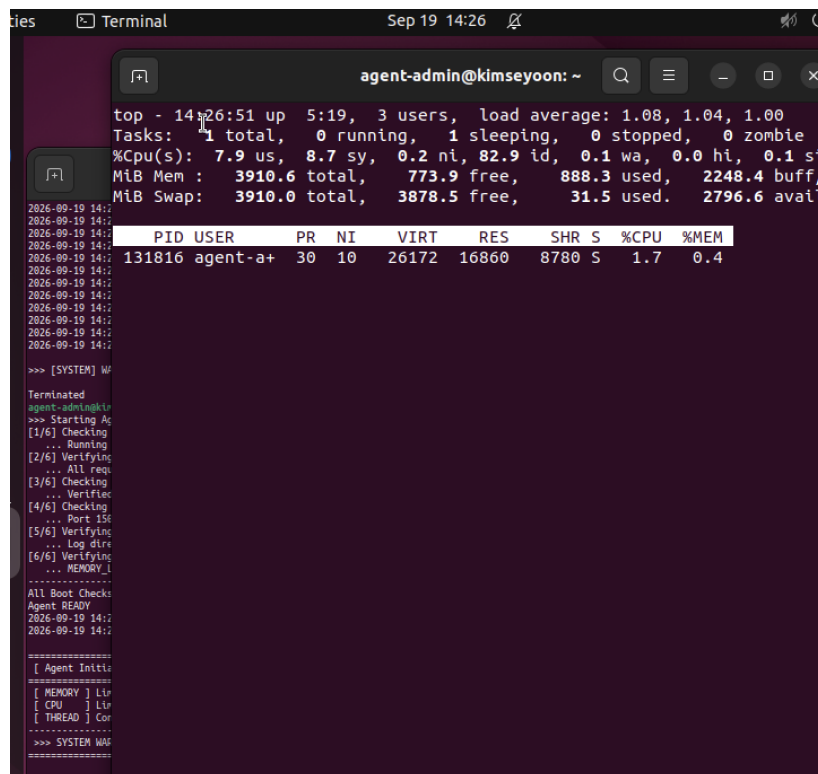
[Bug] CPU Latency - CpuWorker 임계치 위반 후 Watchdog SIGTERM

1. Description (현상 설명)

MEMORY_LIMIT=512MB, CPU_MAX_OCCUPY=80%, MULTI_THREAD_ENABLE=true 조건에서 CpuWorker가 선택되었다. 프로그램 내부 Current Load가 증가한 뒤 약 50%대에서 CPU Threshold Violated가 발생했고 Watchdog이 SIGTERM을 발생시켜 프로세스를 종료했다.

2. Evidence & Logs (증거 자료)

프로그램 로그에서 Current Load 증가와 WATCHDOG: INITIATING EMERGENCY ABORT (SIGTERM)을 확인했다. 동일 실행을 Linux top으로 외부 관측했을 때 대상 프로세스의 %CPU는 1.7%로 나타났다. 따라서 프로그램 내부 Current Load와 Linux가 보고하는 프로세스 %CPU를 동일한 지표로 간주하지 않았다.



```
top - 14:26:51 up 5:19, 3 users, load average: 1.08, 1.04, 1.00
Tasks: 1 total, 0 running, 1 sleeping, 0 stopped, 0 zombie
%Cpu(s): 7.9 us, 8.7 sy, 0.2 ni, 82.9 id, 0.1 wa, 0.0 hi, 0.1 si
MiB Mem : 3910.6 total, 773.9 free, 888.3 used, 2248.4 buff/
MiB Swap: 3910.0 total, 3878.5 free, 31.5 used. 2796.6 avail

  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM
 131816 agent-a+   30   10 26172 16860 8780 S   1.7   0.4

>>> [SYSTEM] W
Terminated
agent-admin@kimseyoon:~
>>> Starting Ag
[1/6] Checking
... Running
[2/6] Verifying
... All req
[3/6] Checking
... Verifie
[4/6] Checking
... Port 15
[5/6] Verifying
... Log dir
[6/6] Verifying
... MEMORY I
-----
All Boot Checks
Agent READY
2026-09-19 14:2
2026-09-19 14:2
[ Agent Init: ]
[ MEMORY ] Lit
[ CPU ] Lit
[ THREAD ] Co
>>> SYSTEM WAF
-----
```

그림 4. Linux top 외부 관측: 대상 프로세스의 실제 %CPU는 1.7%로 확인

```

>>> Starting Agent Boot Sequence...
[1/6] Checking User Account [OK]
... Running as service user 'agent-admin' (uid=1001)
[2/6] Verifying Environment Variables [OK]
... All required Envs correct
[3/6] Checking Required Files [OK]
... Verified 'secret.key' with correct key string.
[4/6] Checking Port Availability [OK]
... Port 15034 is available.
[5/6] Verifying Log Permission [OK]
... Log directory is writable: /var/log/agent-app
[6/6] Verifying Mission Environment [OK]
... MEMORY_LIMIT=512MB, CPU_MAX_OCCUPY=80%, MULTI_THREAD_ENABLE=True
-----
All Boot Checks Passed!
Agent READY
2026-09-19 13:49:38,131 [INFO] [SafetyGuard] Process priority lowered (nice=10).
2026-09-19 13:49:38,131 [INFO] Agent listening at port 15034

=====
[ Agent Initiate ] Resource Check
=====
Terminal Limit: 512MB [ OK ]
          Limit: 80% [ WARNING: Recommend Under 50% ]
[ InKcW ] Concurrency: True [ WARNING ]
-----
>>> SYSTEM WARNING: POTENTIAL DEADLOCK IN CONCURRENT MODE.
=====
2026-09-19 13:49:40,135 [INFO] [CpuWorker] Started. Maximum CPU Limit: 80%
2026-09-19 13:49:40,135 [INFO] [CpuWorker] Current Load: 5.00%
2026-09-19 13:49:43,257 [INFO] [CpuWorker] Current Load: 5.52%
2026-09-19 13:49:46,376 [INFO] [CpuWorker] Current Load: 13.75%
2026-09-19 13:49:49,496 [INFO] [CpuWorker] Current Load: 21.47%
2026-09-19 13:49:52,621 [INFO] [CpuWorker] Current Load: 29.69%
2026-09-19 13:49:55,744 [INFO] [CpuWorker] Current Load: 32.76%
2026-09-19 13:49:58,869 [INFO] [CpuWorker] Current Load: 39.62%
2026-09-19 13:50:01,993 [INFO] [CpuWorker] Current Load: 47.54%
2026-09-19 13:50:05,109 [INFO] [CpuWorker] Current Load: 54.37%
2026-09-19 13:50:05,214 [CRITICAL] [CpuWorker] CPU Threshold Violated! (54.37%).

>>> [SYSTEM] WATCHDOG: INITIATING EMERGENCY ABORT (SIGTERM) <<<
Terminated

```

그림 5. CPU 장애 실행 로그: CpuWorker 부하 증가 및 Watchdog에 의한 종료 과정

3. Root Cause Analysis (원인 분석)

직접적인 종료 원인은 CpuWorker가 내부 부하 판정값을 임계 상태로 판단하여 Watchdog 보호 로직을 실행한 것이다. Linux top/ps 관측치는 1%대였으므로 내부 Current Load의 계산 방식은 OS의 %CPU와 다르다고 판단했다. 제공된 실행 파일의 내부 계산 방식을 역공학하지 않았으며, 관측된 두 지표의 차이를 그대로 기록했다. 일반적으로 CPU를 장시간 과도하게 점유하는 작업은 다른 프로세스의 스케줄링 기회를 줄여 지연을 증가시킬 수 있다.

4. Workaround & Verification (조치 및 검증)

CPU_MAX_OCCUPY를 80%에서 40%로 변경했다. 최종 정상 설정에서는 CPU 항목이 [OK]로 판정되었고, CpuWorker가 40.00%에 도달했을 때 종료 대신 'Starting cooldown'으로 전환된 뒤 부하가 감소하고 다시 작업을 이어갔다. 따라서 80% 조건의 Watchdog SIGTERM과 달리 40% 조건에서는 CPU 관련 강제 종료가 발생하지 않았다.

```
agent-admin@kinseyoon:/agent-app$ export MULTI_THREAD_ENABLE=false
./agent-leak-app-arm64
>>> Starting Agent Boot Sequence...
[1/6] Checking User Account [OK]
... Running as service user 'agent-admin' (uid=1001)
[2/6] Verifying Environment Variables [OK]
... All required Envs correct
[3/6] Checking Required Files [OK]
... Verified 'secret.key' with correct key string.
[4/6] Checking Port Availability [OK]
... Port 15034 is available.
[5/6] Verifying Log Permission [OK]
... Log directory is writable: /var/log/agent-app
[6/6] Verifying Mission Environment [OK]
... MEMORY_LIMIT=512MB, CPU_MAX_OCCUPY=40%, MULTI_THREAD_ENABLE=False
-----
All Boot Checks Passed!
Agent READY
2026-09-19 14:35:05,482 [INFO] [SafetyGuard] Process priority lowered (nice=10).
2026-09-19 14:35:05,482 [INFO] Agent listening at port 15034

=====
[ Agent Initiate ] Resource Check
=====
[ MEMORY ] Limit: 512MB [ OK ]
[ CPU ] Limit: 40% [ OK ]
[ THREAD ] Concurrency: False [ OK ]
=====
>>> SYSTEM STATUS: STABLE. STARTING WORKLOAD MONITORING...
=====

2026-09-19 14:35:07,493 [INFO] >>> Scenario Selected: [Healthy System Monitoring]

>>> [SYSTEM] ALL CONFIGURATIONS OPTIMAL. RUNNING STABILITY TEST... <<<

2026-09-19 14:35:07,493 [INFO] [Scheduler] Task Scheduler Initialized.
2026-09-19 14:35:07,493 [INFO] [Scheduler] Registered Tasks: ['Thread-A', 'Thread-B', 'Thread-C']
2026-09-19 14:35:07,493 [INFO] [Scheduler] Starting task execution...
2026-09-19 14:35:07,493 [INFO] [Thread-A] Task Started. Calculating... (20%)
2026-09-19 14:35:07,546 [INFO] [Thread-A] Calculating... (40%)
2026-09-19 14:35:07,597 [INFO] [Thread-A] Preempted. Progress saved at (40%)
2026-09-19 14:35:07,648 [INFO] [Thread-B] Task Started. Calculating... (20%)
2026-09-19 14:35:07,699 [INFO] [Thread-B] Calculating... (40%)
2026-09-19 14:35:07,753 [INFO] [Thread-B] Preempted. Progress saved at (40%)
2026-09-19 14:35:07,804 [INFO] [Thread-C] Task Started. Calculating... (20%)
2026-09-19 14:35:07,855 [INFO] [Thread-C] Calculating... (40%)
2026-09-19 14:35:07,906 [INFO] [Thread-C] Preempted. Progress saved at (40%)
2026-09-19 14:35:07,958 [INFO] [Thread-A] Resumed. Calculating... (60%)
2026-09-19 14:35:08,013 [INFO] [Thread-A] Calculating... (80%)
2026-09-19 14:35:08,065 [INFO] [Thread-A] Preempted. Progress saved at (80%)
2026-09-19 14:35:08,118 [INFO] [Thread-B] Resumed. Calculating... (60%)
2026-09-19 14:35:08,169 [INFO] [Thread-B] Calculating... (80%)
2026-09-19 14:35:08,221 [INFO] [Thread-B] Preempted. Progress saved at (80%)
2026-09-19 14:35:08,275 [INFO] [Thread-C] Resumed. Calculating... (60%)
2026-09-19 14:35:08,328 [INFO] [Thread-C] Calculating... (80%)
2026-09-19 14:35:08,380 [INFO] [Thread-C] Preempted. Progress saved at (80%)
2026-09-19 14:35:08,431 [INFO] [Thread-A] Resumed. Calculating... (100%)
2026-09-19 14:35:08,483 [INFO] [Thread-B] Resumed. Calculating... (100%)
```

그림 6. 최종 정상 설정(512MB / 40% / false): CPU 40% [OK], Healthy System Monitoring 및 정상 작업 진행

[Bug] Deadlock - 두 작업 스레드의 상호 Lock 대기

1. Description (현상 설명)

MEMORY_LIMIT=512MB, CPU_MAX_OCCUPY=40%, MULTI_THREAD_ENABLE=true 조건에서 동시 처리 스레드가 실행되었다. Worker - Thread - 1은 Shared_Memory_A를 획득한 뒤 Socket_Pool_B를 기다렸고, Worker - Thread - 2는 Socket_Pool_B를 획득한 뒤 Shared_Memory_A를 기다렸다. 두 스레드가 모두 BLOCKED 상태가 된 이후 로그 진행이 멈췄다.

2. Evidence & Logs (증거 자료)

프로그램 로그에 각 스레드의 LOCK ACQUIRED와 상대 자원에 대한 WAITING/BLOCKED가 연속으로 기록되었다. ps -ef로 부모/자식 프로세스가 종료되지 않고 존재함을 확인했고, ps -l로 자식 프로세스 내부에 메인 스레드와 두 작업 스레드가 존재하며 모두 %CPU 0.0 상태임을 확인했다.

```
agent-admin@kimseoyoon:~/agent-app$ export CPU_MAX_OCCUPY=40
./agent-leak-app-arm64
>>> Starting Agent Boot Sequence...
[1/6] Checking User Account [OK]
... Running as service user 'agent-admin' (uid=1001)
[2/6] Verifying Environment Variables [OK]
... All required Envs correct
[3/6] Checking Required Files [OK]
... Verified 'secret.key' with correct key string.
[4/6] Checking Port Availability [OK]
... Port 15034 is available.
[5/6] Verifying Log Permission [OK]
... Log directory is writable: /var/log/agent-app
[6/6] Verifying Mission Environment [OK]
... MEMORY_LIMIT=512MB, CPU_MAX_OCCUPY=40%, MULTI_THREAD_ENABLE=True
-----
All Boot Checks Passed!
Agent READY
2026-09-19 14:28:27,444 [INFO] [SafetyGuard] Process priority lowered (nice=10).
2026-09-19 14:28:27,444 [INFO] Agent listening at port 15034

=====
[ Agent Initiate ] Resource Check
=====
[ MEMORY ] Limit: 512MB [ OK ]
[ CPU ] Limit: 40% [ OK ]
[ THREAD ] Concurrency: True [ WARNING ]
=====
>>> SYSTEM WARNING: POTENTIAL DEADLOCK IN CONCURRENT MODE.
=====

2026-09-19 14:28:29,455 [WARNING] [AgentWorker] Initializing concurrent transaction processors...
2026-09-19 14:28:29,455 [WARNING] [System] CAUTION: Strict resource locking is enabled.
2026-09-19 14:28:34,482 [INFO] [Worker-Thread-1] Process Started. Attempting to lock [Shared_Memory_A]...
2026-09-19 14:28:34,482 [INFO] [AgentWorker][Worker-Thread-1] LOCK ACQUIRED: [Shared_Memory_A]. (Holding...)
2026-09-19 14:28:34,482 [INFO] [AgentWorker][Worker-Thread-1] Processing critical data in Memory A...
2026-09-19 14:28:34,482 [INFO] [AgentWorker][Worker-Thread-2] Process Started. Attempting to lock [Socket_Pool_B]...
2026-09-19 14:28:34,482 [INFO] [AgentWorker][Worker-Thread-2] LOCK ACQUIRED: [Socket_Pool_B]. (Holding...)
2026-09-19 14:28:34,482 [INFO] [AgentWorker][Worker-Thread-2] Establishing network connections in Pool B...
2026-09-19 14:28:34,482 [INFO] [AgentWorker] Waiting for worker threads to complete transactions...
2026-09-19 14:28:36,493 [INFO] [AgentWorker][Worker-Thread-2] Need resource [Shared_Memory_A] to write logs.
2026-09-19 14:28:36,493 [INFO] [AgentWorker][Worker-Thread-2] WAITING for [Shared_Memory_A]... (Status: BLOCKED)
2026-09-19 14:28:36,493 [INFO] [AgentWorker][Worker-Thread-1] Need resource [Socket_Pool_B] to finish job.
2026-09-19 14:28:36,493 [INFO] [AgentWorker][Worker-Thread-1] WAITING for [Socket_Pool_B]... (Status: BLOCKED)
```

그림 7. Deadlock 발생: 각 스레드가 서로 다른 Lock을 보유한 채 상대 자원을 WAITING/BLOCKED

```
agent-admin@kimseyoon:~$ ps -ef | grep '[a]gent-leak-app-arm64'
agent-a+  131892    3908  0 14:28 pts/1    00:00:00 ./agent-leak-app-arm64
agent-a+  131893  131892  0 14:28 pts/1    00:00:00 ./agent-leak-app-arm64
agent-admin@kimseyoon:~$
```

그림 8. ps -ef: 장애 중에도 부모 PID 131892와 자식 PID 131893이 살아 있음

```
agent-admin@kimseyoon:~$ ps -L -p 131893 -o pid,tid,ppid,pcpu,pmem,rss,stat,comm
  PID      TID      PPID %CPU %MEM   RSS STAT COMMAND
 131893  131893   131892  0.0  0.4 16852 SNl+ agent-leak-app-
 131893  131895   131892  0.0  0.4 16852 SNl+ agent-leak-app-
 131893  131896   131892  0.0  0.4 16852 SNl+ agent-leak-app-
agent-admin@kimseyoon:~$
```

그림 9. ps -L: 자식 프로세스 내 3개 TID 확인, 각 스레드 %CPU 0.0으로 진행 정체

3. Root Cause Analysis (원인 분석)

Thread - 1은 자원 A를 보유한 상태에서 B를 기다리고, Thread - 2는 자원 B를 보유한 상태에서 A를 기다려 순환 대기가 형성되었다. 상호배제, 점유하며 대기, 비선점, 순환 대기의 네 가지 Coffman 조건이 모두 성립하는 전형적인 Deadlock 구조이다. 프로세스 자체는 살아 있지만 두 작업 스레드가 서로의 Lock 해제를 기다리므로 유의미한 작업 진행이 중단된다.

4. Workaround & Verification (조치 및 검증)

MULTI_THREAD_ENABLE을 true에서 false로 변경했다. 최종 실행에서는 THREAD Concurrency가 False [OK]로 표시되고 Healthy System Monitoring 시나리오가 선택되었다. Thread - A/B/C가 정상적으로 실행 · 재개 · 완료되었으며 이후 CPU cooldown과 메모리 cleanup도 정상 수행되어 프로세스가 지속 실행되었다.

```

agent-admin@tkinseyoon:~/agent-app$ export MULTI_THREAD_ENABLE=false
./agent-leak-app-arm64
>>> Starting Agent Boot Sequence...
[1/6] Checking User Account [OK]
... Running as service user 'agent-admin' (uid=1001)
[2/6] Verifying Environment Variables [OK]
... All required Envs correct
[3/6] Checking Required Files [OK]
... Verified 'secret.key' with correct key string.
[4/6] Checking Port Availability [OK]
... Port 15034 is available.
[5/6] Verifying Log Permission [OK]
... Log directory is writable: /var/log/agent-app
[6/6] Verifying Mission Environment [OK]
... MEMORY_LIMIT=512MB, CPU_MAX_OCCUPY=40%, MULTI_THREAD_ENABLE=False
-----
All Boot Checks Passed!
Agent READY
2026-09-19 14:35:05,482 [INFO] [SafetyGuard] Process priority lowered (nice=10).
2026-09-19 14:35:05,482 [INFO] Agent listening at port 15034
=====
[ Agent Initiate ] Resource Check
=====
[ MEMORY ] Limit: 512MB [ OK ]
[ CPU ] Limit: 40% [ OK ]
[ THREAD ] Concurrency: False [ OK ]
=====
>>> SYSTEM STATUS: STABLE. STARTING WORKLOAD MONITORING...
=====
2026-09-19 14:35:07,493 [INFO] >>> Scenario Selected: [Healthy System Monitoring]

>>> [SYSTEM] ALL CONFIGURATIONS OPTIMAL. RUNNING STABILITY TEST... <<<

2026-09-19 14:35:07,493 [INFO] [Scheduler] Task Scheduler Initialized.
2026-09-19 14:35:07,493 [INFO] [Scheduler] Registered Tasks: ['Thread-A', 'Thread-B', 'Thread-C']
2026-09-19 14:35:07,493 [INFO] [Scheduler] Starting task execution...
2026-09-19 14:35:07,493 [INFO] [Thread-A] Task Started. Calculating... (20%)
2026-09-19 14:35:07,546 [INFO] [Thread-A] Calculating... (40%)
2026-09-19 14:35:07,597 [INFO] [Thread-A] Preempted. Progress saved at (40%)
2026-09-19 14:35:07,648 [INFO] [Thread-B] Task Started. Calculating... (20%)
2026-09-19 14:35:07,699 [INFO] [Thread-B] Calculating... (40%)
2026-09-19 14:35:07,753 [INFO] [Thread-B] Preempted. Progress saved at (40%)
2026-09-19 14:35:07,804 [INFO] [Thread-C] Task Started. Calculating... (20%)
2026-09-19 14:35:07,855 [INFO] [Thread-C] Calculating... (40%)
2026-09-19 14:35:07,906 [INFO] [Thread-C] Preempted. Progress saved at (40%)
2026-09-19 14:35:07,958 [INFO] [Thread-A] Resumed. Calculating... (60%)
2026-09-19 14:35:08,013 [INFO] [Thread-A] Calculating... (80%)
2026-09-19 14:35:08,065 [INFO] [Thread-A] Preempted. Progress saved at (80%)
2026-09-19 14:35:08,118 [INFO] [Thread-B] Resumed. Calculating... (60%)
2026-09-19 14:35:08,169 [INFO] [Thread-B] Calculating... (80%)
2026-09-19 14:35:08,221 [INFO] [Thread-B] Preempted. Progress saved at (80%)
2026-09-19 14:35:08,275 [INFO] [Thread-C] Resumed. Calculating... (60%)
2026-09-19 14:35:08,328 [INFO] [Thread-C] Calculating... (80%)
2026-09-19 14:35:08,380 [INFO] [Thread-C] Preempted. Progress saved at (80%)
2026-09-19 14:35:08,431 [INFO] [Thread-A] Resumed. Calculating... (100%)
2026-09-19 14:35:08,483 [INFO] [Thread-B] Resumed. Calculating... (100%)

```

그림 10. MULTI_THREAD_ENABLE=false: 모든 설정 [OK], 스레드 작업이 BLOCKED 없이 정상 진행

최종 검증 요약

항목	Before	조치	After
OOM	256MB → 275MB에서 종료	MEMORY_LIMIT=512MB	525MB 도달 후 cleanup / 계속 실행
CPU	80% → Watchdog SIGTERM	CPU_MAX_OCCUPY=40%	Peak 후 cooldown / 계속 실행
Deadlock	true → WAITING/BLOCKED	MULTI_THREAD_ENABLE=false	정상 스레드 진행 / 계속 실행

최종 정상 설정은 MEMORY_LIMIT=512MB, CPU_MAX_OCCUPY=40%, MULTI_THREAD_ENABLE=false이다. 이 조건에서 프로그램은 Healthy System Monitoring 상태로 진입했으며 CPU 부하는 cooldown으로 제어되고, 메모리 한계 도달 시 cleanup과 cache flush를 수행한 뒤 계속 실행되었다. Deadlock을 유발했던 동시 Lock 대기 역시 재현되지 않았다.